
Not All Spines Are Created Equal: How CT Segmenters Fail on Lumbosacral Transitional Vertebrae

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Abstract

Wrong-level spine surgery remains a persistent never-event, and the lumbosacral transitional vertebra (LSTV, 5–35% prevalence) is its largest anatomic driver. We audit TotalSegmentator—the most widely deployed CT segmentation system and default back-end of 3D Slicer—and show that on lumbarization cases its output reproduces the pattern of a one-level miscount: lacking an L6 class, it shifts lumbar labels caudally by one level, and junction-DSC drops 29 points on Any-LSTV ($0.824 \rightarrow 0.534$; $n=7$ cases with a scorable junction). No existing CT benchmark surfaces this—not TotalSegmentator’s own evaluation, not Li et al. [2025] (the per-class DSC frontier on the matched COLONOG split), and not VERIDAH (vertebra-labeling SOTA on a private cohort, on pre-localized crops with no pelvis). We contribute: (i) an LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) that measures the level shift and frames it as a plausible automated wrong-level-risk mechanism; (ii) CTSPINOPELVIC1K—1,153 CT records across 802 patients with unified spinopelvic masks, 33 LSTV-positive patients in a six-way phenotype taxonomy derived from the source annotations’ filenames and lumbar counts, and Castellvi typing (I–IV) of all 33 by two radiology residents ($\kappa=0.74$); (iii) the first publicly released merge-based dual-spinopelvic CT segmenter, exceeding TotalSegmentator on sacrum ($0.817 \rightarrow 0.969$) with released weights; (iv) a same-data, same-fold unmerged control with an explicit L6 class that still labels most ground-truth L6 voxels L5, isolating the training-side L5/L6 merge as the component that removes the collision, while a residual L4/last_lumbar collision on count-style sacralization motivates a downstream sequence predictor. Only 7 LSTV cases fall in the held-out split; every stratified number is reported with its n .

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1 Introduction

Wrong-level spine surgery is a persistent never-event in operative spine care [Konin and Walz, 2010]. Despite intraoperative imaging and surgical-pause checklists, the most common cause—failure to correctly count vertebral levels in the presence of an LSTV—continues to account for the majority of reported wrong-level lumbar incidents [Carrino et al., 2011]. The mechanism is well documented: a surgeon counts lumbar bodies upward from the sacrum, finds five (because lumbarization presents a sixth lumbar body that has not been recognized), and labels the bottom body as L5 when it is anatomically L6. Surgery proceeds one level caudal to the intended target.

Modern radiology workflows increasingly route automated segmentation as an upstream substrate for clinical decision-making. The most widely deployed CT segmentation system in this role is TotalSegmentator [Wasserthal et al., 2023]: an nnU-Net trained on 1,204 routine CTs to delineate 104 anatomic structures, the default segmentation back-end of 3D Slicer, with thousands of clinical and research installations. The original paper reports an aggregate test-set Dice of 0.943 across all 104 classes and qualitatively documents two spine failure modes affecting neighboring-vertebrae label confusion (1% of cases) and neighboring-rib confusion (12%). TotalSegmentator is neither the only public CT spine segmenter nor the per-class DSC frontier on the matched CTSpine1K + CTPelvic1K (COLONOG) split (§2); it is, however, the model surgeons and downstream tools consume. This paper concerns one anatomic class on which TotalSegmentator does not perform at its headline level, and the consequences when it does not.

The risk mechanism. On lumbarized spines, TotalSegmentator produces an output pattern that matches the mechanism of the surgical miscount. Its label vocabulary contains L1–L5 but no L6 class. On a lumbarized spine (six lumbar bodies plus sacrum), the model distributes its five lumbar labels across the six bodies by shifting caudally by one anatomic level: the body that is anatomically L6 receives the L5 label, the body that is anatomically L5 receives the L4 label, and so on. A plan that took the disc between predicted-L4 and predicted-L5 as L4–L5 would land on the anatomical L5–L6 disc. We present this as a *plausible automated wrong-level-risk mechanism*: the paper measures how often the shifted output occurs and what it looks like at the junction. It does not evaluate surgical-planning software, reader behaviour or downstream workflow, and no wrong-level event is attributed to a segmenter here.

The architectural alternative: VERIDAH. Möller et al. [2026] label transitional vertebrae with a per-vertebra classifier trained with L5/L6 merged, followed by a constrained sequence predictor that restores the count (§2). Whether a voxel-wise network trained at full resolution with the same merge needs the sequence predictor at all has never been tested; we supply that test.

The evaluation gap. Neither TotalSegmentator, VerSe [Sekuboyina et al., 2021], CTSpine1K [Deng et al., 2021], nor CTPelvic1K [Liu et al., 2021] report LSTV-stratified performance; earlier CT labelers excluded the failure mode [Sekuboyina et al., 2020], and VERIDAH reports anomaly performance on a private cohort. A production model therefore emits a confident, locally well-formed, caudally-shifted prediction on a 5–35%-prevalence variant with no error signal at evaluation time. Closing the gap requires hardware: fitting lumbar spine and pelvis at $0.75 \times 0.75 \times 0.80$ mm and a $256 \times 320 \times 320$ patch consumes ~ 100 GB of GPU memory, available on an H200 but not on the L40S/A100/H100 GPUs that bracket TotalSegmentator’s training configuration.

Contributions.

- (i) **The level shift in TS, quantified.** An LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) measures TS’s systematic one-level caudal shift on lumbarization (L4=0.388, L5=0.291 over 14 patients; 95%+ of GT-L6 voxels labeled L5 on the case study, §4) and the structural absence of L6 across all evaluation cases, and frames the shift as a plausible automated wrong-level-risk mechanism.
- (ii) **CTSPINOPELVIC1K: a stratified spinopelvic benchmark.** 1,153 abdominal CT records across 802 patients, unified spinopelvic masks, 33 LSTV-positive patients in a six-way subtype taxonomy (lumb $n=14$; sac_{count} $n=9$; full sacralization $n=6$; semi-sacralization

$n=2$; ambiguous $n=2$), and a registration-free fusion pipeline that resolves a previously undocumented Y-axis coordinate-convention mismatch between CTSpine1K and CTPelvic1K. The subtypes come from the source annotations (filename qualifiers and lumbar counts), not from image reads; the Castellvi layer (I–IV, all 33) is a two-resident read with $\kappa=0.74$ and no senior adjudication (§5). Both source datasets share the COLONOG cohort that underlies VerSe and CTSpine1K, but neither encodes pelvic context with LSTV stratification.

- (iii) **First publicly released merge-based LSTV-handling dual-spinopelvic CT segmenter.** A 5-fold CV nnU-Net 3d_fullres baseline on H200 hardware adopting VERIDAH’s training-side merge (last_lumbar = L5+L6) at the voxel level, with pelvic-class supervision in a single forward pass. We exceed TS on sacrum and uniquely deliver both spine and pelvis with released weights (Table 5); §2 details the reproducibility tiers separating openly-released from paper-only baselines.
- (iv) **Empirical isolation: the merge is necessary and insufficient.** A same-data, same-fold, same-architecture control trained *without* the merge, with an explicit L6 class and L6 oversampling, still labels most GT-L6 voxels L5 (§8.1), so the merge—not the added data or the added class—removes the L5/L6 collision driving TS’s level shift. The merge in turn exposes a residual L4/last_lumbar collision on count-style sacralization, isolating the component of VERIDAH’s success attributable to its sequence predictor—a direction VERIDAH’s authors could not ablate.

2 Related Work

Wrong-level spine surgery and LSTV. [Konin and Walz, 2010, Carrino et al., 2011, Sekharappa et al., 2014] document LSTV-induced miscounting (5–35% prevalence) without addressing automated segmentation as a miscount-propagation pathway. Kwak et al. [2025] detect LSTV on plain radiographs (binary; different modality).

Production CT spine and pelvic segmentation. TotalSegmentator [Wasserthal et al., 2023], an nnU-Net trained on 1,204 CTs and the default 3D Slicer back-end, has no L6 class and no LSTV stratification; we audit it as the production system downstream pipelines actually consume, not as a per-class DSC frontier. Li et al. [2025] hold the published per-class DSC frontier on a COLONOG split (731 scans; 440/146/145) but release neither weights nor code (paper-only) and do not stratify by LSTV. Liu et al. [2025] release FracSegNet (PENGWIN+CTPelvic1K), a pelvic-only cascaded 3D nnU-Net providing sacrum and L/R hipbone; their qualitative note of sacralization degradation (their §4.3) corroborates the LSTV gap our benchmark surfaces. Table 5 compares all baselines.

Vertebra-labeling SOTA: VERIDAH. Möller et al. [2026] address transitional-vertebra *labeling* (distinct from segmentation): a four-head 3D DenseNet169 classifier on per-vertebra crops, trained with L5/L6 (and T12/T13) merged, plus a constrained sequence predictor enforcing 4–6 lumbar bodies (terminal double-L5 encodes L6). On a private CT cohort of 1,536 volumes VERIDAH reports 99.18% subject-level perfect labeling versus prior CT SOTA [Meng et al., 2023] 77.81%. It requires upstream centroids (VerSe-derived models or SPINEPS [Möller et al., 2024]), excludes the pelvis, and validates privately. Its ablation (Table 5) shows the sequence predictor adds substantial value, but the comparison against direct semantic segmentation on a public LSTV-stratified split is missing—the gap our baseline (§7) fills.

Benchmarks and evaluation rigor. VerSe [Sekuboyina et al., 2021] (460 CT volumes) is the only public CT benchmark whose label vocabulary includes L6 (label 25) but it is unstratified by LSTV and lacks pelvic GT; earlier CT labelers explicitly excluded the failure mode [Sekuboyina et al., 2020]. Stratified safety-critical reporting [Maier-Hein et al., 2024] motivates §6.

3 Failure-Mode Analysis: TS on LSTV

We evaluate TotalSegmentator v2 (total task) zero-shot at full resolution on the full CT-SPINOPELVIC1K matched cohort. TotalSegmentator was not trained on any CTSPINOPELVIC1K case, so every case is valid; this provides 1,152 of CTSPINOPELVIC1K’s 1,153 records (one record dropped due to TS inference failure) aggregated to 801 patients via patient-level merging, including

Table 1: TS zero-shot stratified by LSTV phenotype on CTSPINOPELVIC1K, full-resolution mode; n is patients. Junction DSC over a 40 mm L5/S1 axial window is scorable only where GT holds L5 and sacrum (n_{jxn}). †: no record with a scorable junction.

Subgroup	n	L1	L2	L3	L4	L5	Sac	Hips	Jxn DSC	n_{jxn}
All cases	801	0.911	0.906	0.913	0.914	0.913	0.817	0.929	0.818	341
Normal	768	0.921	0.918	0.929	0.936	0.938	0.817	0.929	0.824	334
Any LSTV	33	0.681	0.615	0.523	0.367	0.314	0.822	0.929	0.534	7
Lumbarization	14	0.878	0.853	0.690	0.388	0.291	0.833	0.928	0.357	3
Sacralization (count-style)	9	0.265	0.100	0.055	0.001	0.000	0.814	0.926	0.697	2
Semi-sacralization	2	0.951	0.951	0.951	0.955	0.956	0.824	0.937	†	0
Sacralization (full)	6	0.718	0.676	0.673	0.724	0.793	0.826	0.932	0.636	2
Ambiguous	2	0.952	0.951	0.882	0.678	0.374	0.771	0.932	†	0
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33 LSTV-positive patients. Full-resolution mode isolates the failure as a property of label vocabulary and training distribution rather than inference resolution.

Table 1 reports performance by phenotype. On normal anatomy ($n=768$) TotalSegmentator performs at its published level. On Any-LSTV ($n=33$) every lumbar metric drops; lumbarization collapses caudally (L1=0.878 $\rightarrow$ L4=0.388 $\rightarrow$ L5=0.291 over 14 patients). Junction-DSC is scorable only where the ground truth holds both L5 and sacrum: 334 normal patients and 7 LSTV patients (3 lumbarization: 0.22, 0.23, 0.62; 2 count-style sacralization: 0.47, 0.93; 2 full sacralization: 0.43, 0.84), so the 29-point Any-LSTV drop in the abstract rests on 7 patients. Per patient, L5 DSC falls below 0.5 in 10 of 14 lumbarization cases (Wilson 95% interval 45–88%) and in 9 of 9 count-style sacralization cases (70–100%), against 9 of 750 normal cases (0.6–2.3%). L6 predictions are structurally zero across all 1,152 records (0–0.3%).

The level-shift mechanism. The asymmetry between lumbarization (severe, caudally progressive) and count-style sacralization (uniform collapse) follows from a single fact: TotalSegmentator’s output head has no L6 class. For 6-vertebra anatomies the model compresses 6 anatomic units into 5 labels by shifting label assignment caudally; the most rostral body remains L1 (escape into thoracic vertebrae), but every successive body is misaligned, with the final body labeled L5 when it is anatomically L6. For 4-vertebra anatomies the model emits standard L1–L5, overshooting the GT lumbar count and producing a +1-level offset uniformly. The level-shift mechanism is the segmentation analog of the surgical miscount [Konin and Walz, 2010].

4 Case Study: the Level Shift at the Junction

Aggregate metrics establish the population pattern; the case study establishes the mechanism at the level a surgeon would consume the output. Token 215 is a lumbarization case (six lumbar bodies; Castellvi Ib by consensus, one of the six cases the two readers graded differently, §5). Table 2 reports per-class DSC: L1 and L2 are well-placed (0.954, 0.927), L3 is partially correct (0.594), and **L4, L5, and L6 all collapse to dice zero**—no GT-L4 or GT-L5 voxel correctly labeled and no L6 voxel emitted. The pelvic classes remain at near-Normal accuracy: the failure is local to the lumbosacral count, not a general degradation of TS on this patient.

Table 2: Token 215 (lumbarization, 6 lumbar bodies): per-class TS zero-shot DSC. L4, L5 and L6 collapse to zero because TS compresses 6 bodies into 5 labels; pelvic classes are unaffected. † TS has no L6 label.

Class	L1	L2	L3	L4	L5	L6	Sacrum	Hip _L	Hip _R
DSC	0.954	0.927	0.594	0.000	0.000	0.000 †	0.817	0.922	0.926

Why the pattern matters. A reader who took the segmentation at face value and targeted the disc between predicted-L4 (anatomical L5) and predicted-L5 (anatomical L6) would be one level caudal to an L4–L5 intent. The output is locally well-formed: five compact vertebral predictions in roughly

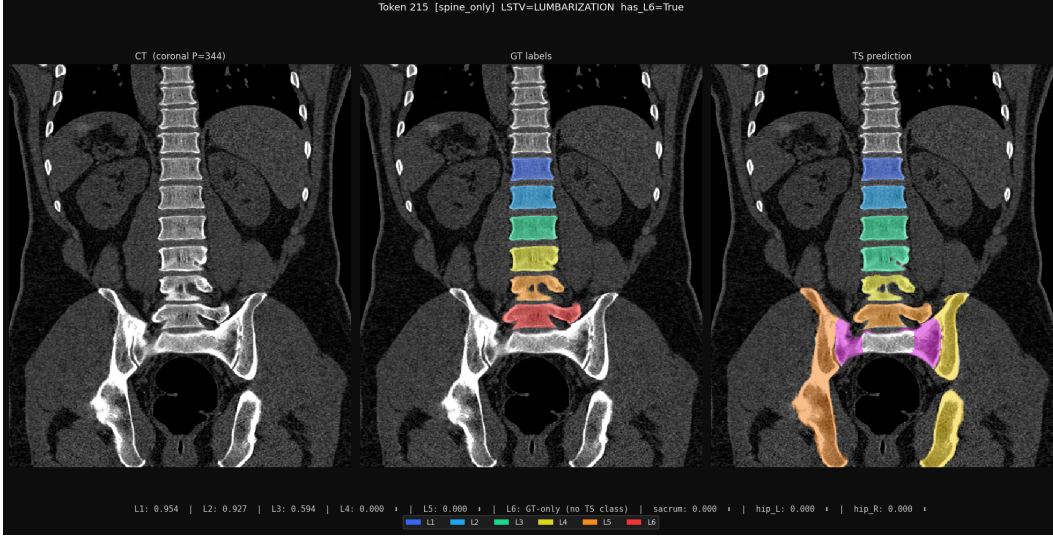


Figure 1: Token 215, lumbarization (6 lumbar bodies): TS predictions on CT. Predicted-L3 absorbs anatomical L3 and L4, predicted-L4 sits on anatomical L5, predicted-L5 on anatomical L6, which has no label in TS’s output head. Five compact, well-formed predictions, every label below L2 one level off. Quantified in Table 2.

correct positions, detectable only by counting bodies (six lumbar, not five), which downstream pipelines do not perform. Whether a surgeon or a planning tool would in fact act on such an output is not tested here; the claim is the mechanism and its frequency, not a measured clinical harm.

5 CTSPINOPELVIC1K: Construction and Composition

CTSPINOPELVIC1K fuses CT scans from COLONOG [Smith et al., 2015] with annotations from CTSpine1K [Deng et al., 2021] (VerSe-derived per-vertebra masks) and CTPelvic1K [Liu et al., 2021] (sacrum and bilateral hips). The same COLONOG cohort underlies VerSe and CTSpine1K; CTSPINOPELVIC1K’s contribution is the patient-level fusion of CTSpine1K’s vertebra masks with CTPelvic1K’s pelvic masks plus the LSTV-stratification overlay neither source provides. Of 1,153 matched records, 342 carry fused (spine + pelvis) ground truth, 352 are separate-mode, 90 are spine-only, and 20 are pelvic-only (Table 3).

Release used. Every number in this paper was computed on the April 2026 release (the HuggingFace export finalized 25 April 2026, later labelled v1 in the repository; training began 3 May 2026): 1,153 records across 802 patients, a ten-class scheme (background, L1–L6, sacrum, left and right hip) merged to nine classes for training, and separate spine-only and pelvic-only records for patients whose two source annotations lie on different acquisitions. The archive of record is the Zenodo concept DOI 10.5281/zenodo.22139642, whose current version is v11 (10.5281/zenodo.22797014, 18 September 2026). v11 differs from the April release as follows: one record per patient (802 records), each separate-mode patient’s missing pelvis completed on the spine’s series by out-of-fold pseudolabelling; VerSe-native identifiers (L1–L6 = 20–25, sacrum 26, hips 30/31) with femora, per-level ribs, lumbar ribs and surgical hardware added as further classes; hip-laterality corrections on 22 records and hand corrections on five; both Castellvi reads and their consensus in the manifest; and the CT images supplied by TCIA series crosswalk and rebuild script, with a HuggingFace mirror carrying the volumes. The 802 patients and the 33 Castellvi-graded patients are the same in both. Nothing here was recomputed on v11.

Patient-anchored fusion and coordinate correction. Patient identity is resolved via canonical UID matching (825/825 patients). Each mask is assigned to its source CT acquisition by intra-patient bone-HU maximization, $s^*(m) = \arg \max_{s \in \mathcal{S}_p} |\{v \in \mathcal{L}(m, s) : I_s(v) > 200 \text{ HU}\}| / |\mathcal{L}(m, s)|$, a metadata-free oracle since SeriesDescription fields are unreliable in TCIA releases (99.6% spine,

Table 3: CTSPINOPELVIC1K composition; LSTV phenotype breakdown and split counts in Table 4.

Property	Value	LSTV cases
Patients matched (CTSPINOPELVIC1K)	802	—
Records matched (CTSPINOPELVIC1K)	1,153	33 patients (44 records)
Fused (spine + pelvis)	342	—
Separate (spine + pelvic)	352	—
Spine-only / Pelvic-only	90 / 20	—

100% pelvic placement). CTSpine1K and CTPelvic1K were converted from DICOM with different `dcm2nix` invocations, producing a systematic Y-axis sign reflection in NIfTI affines; we detect via affine-derived center sign and reverse along axis 1 with affine update. We adopt VERIDAH’s training-side merged-label scheme at the voxel level: 0 = bg, 1–4 = L1–L4, **5 = last_lumbar** (anatomical L5 in 5-vertebra spines; both L5 and L6 in 6-vertebra spines), 6 = sacrum, 7–8 = left/right hip, 9 = ignore.

Six-way LSTV taxonomy, and where its labels come from. No subtype in this benchmark was assigned by reading an image. Each is a deterministic function of the two source annotations: the number of lumbar labels in the CTSpine1K mask (a radiologist’s count, in VerSe convention, so a sixth lumbar body carries VerSe label 25) and the qualifier tag in the CTPelvic1K mask filename (sacralization, semisacralization), whose origin CTPelvic1K does not document. The six subtypes are **normal**, **lumb** (six lumbar labels or label 25 present), **sacr_count** (four lumbar labels and no qualifier), **semisacralization** and **sacralization** (qualifier tag; the lumbar count is five), and **ambiguous** (the two sources disagree). Two of the six therefore rest entirely on a filename tag, and the count subtypes inherit each source’s naming convention for the transitional segment. An earlier substring-match parser collapsed semisacralization into full sacralization—merging two phenotypes whose failure modes diverge (full sacralization preserves the lumbar count and near-normal per-class DSC; semisacralization confuses the L5/S1 boundary); we identify and fix this. Without the corrected taxonomy the qualitatively different failure patterns across LSTV variants (§3) are invisible.

Partial-annotation training protocol. Training only on fused records discards 70% of the data. We use nnU-Net v2’s `ignore_label` [Isensee et al., 2024] to train on all 1,153: voxels in unsupervised regions are written as label 9, contributing zero to Dice and cross-entropy. Spine classes (1–5) receive supervision from 784 records; pelvic classes (6–8) from 714. This is a $3.36\times$ effective expansion. Per-record subtype refinement gates training-time tags on supervised anatomy.

Castellvi annotation layer. The 6-way taxonomy captures the *count* dimension of LSTV. The Castellvi classification [Castellvi et al., 1984] captures the orthogonal *morphological* axis: Type I (dysplastic transverse process), II (pseudoarticulation), III (bony fusion), or IV (mixed II/III), each a (unilateral) or b (bilateral). It is the surgeon’s pre-operative wrong-level-risk indicator, and what a CT segmenter would need to predict to flag LSTV for review rather than silently emit a caudally-shifted label. Two PGY-2 radiology residents each graded all 33 LSTV cases independently, extending the schema—originally sacralization-specific—to score the bony junction uniformly across variants with per-side laterality. They agreed on 27 of 33 (Cohen’s $\kappa=0.74$ on the full grade with laterality, 95% interval 0.55–0.92; $\kappa=0.70$ on type I–IV alone; $n=33$) and resolved the six disagreements by consensus; five of the six cross the II/III boundary (articulation versus bony fusion) and one the I/II boundary. There was no attending or neuroradiologist read, no blinded re-read and no screening of the `normal` cohort, so the grades are a two-trainee consensus and every Castellvi-stratified statement below inherits that limit. Both reads and the consensus are released with the data.

Morphology-stratified failure interpretation. The cross-tabulation (Table 4) surfaces two findings with direct bearing on model behavior. (1) **All 9 count-style sacralization cases are Castellvi IIIB**—bilateral bony fusion, a grade both readers assigned to all nine—and 16/33 (48%) of LSTV overall is IIIB. The residual L4/last_lumbar collision (§8.1) therefore occurs precisely when the discriminating cue is locally observable: bony fusion of the bottom lumbar to sacrum is visually distinct from a normal L5–S1 junction, yet the segmenter still emits last_lumbar for those voxels. Direct semantic segmentation has not learned the Castellvi correlate of the count anomaly even when it is morphologically present in the input—an empirical argument for downstream Castellvi-aware reasoning at the lumbosacral junction (whether VERIDAH-style global count constraints or a dedicated Type III

Table 4: Two-reader consensus Castellvi grade for all 33 LSTV patients of CTSPINOPELVIC1K, by count subtype, with the number of patients in the training pool (5-fold CV) and the held-out test split.

Subtype	Ib	IIa	IIb	IIIa	IIIb	IV	Total	Train	Test
Lumbarization	2	1	3	1	6	1	14	11	3
Sacralization (count-style)	0	0	0	0	9	0	9	8	1
Sacralization (full)	0	1	2	0	1	2	6	4	2
Semi-sacralization	0	2	0	0	0	0	2	2	0
Ambiguous	0	0	1	0	0	1	2	1	1
Total	2	4	6	1	16	4	33	26	7

morphology head). (2) Only 2/33 cases are Type I (population prevalence 14–16% [Sekharappa et al., 2014]), reflecting source-dataset selection bias toward morphologically conspicuous presentations. Every quantitative claim in this paper is therefore scoped to the conspicuous regime (Castellvi II–IV, 31 of 33 cases); the clinically subtler Type I fraction is uncharacterized here.

6 Evaluation Protocol

We define four metrics as the minimum required to attribute LSTV failure modes to the wrong-level-risk mechanism [Maier-Hein et al., 2024]: (1) per-class DSC stratified by 6-way LSTV subgroup, exposing class-level degradation; (2) junction-DSC over a 40 mm L5/S1 axial window, the clinical window for surgical planning; (3) per-class voxel confusion at the L5/S1 junction, which exposes structured failure modes (the level shift) that DSC alone cannot characterize; and (4) for the merged-label baseline, three focused per-epoch confusion blocks anchored on the L4/last_lumbar boundary that drives the residual collision (last_lumbar_GT_on_lumb, L4_GT_on_sacr_count, pred_last_lumbar_on_sacr_count) plus a specificity number (last_lumbar_specificity_on_sacr_count). Every stratified number is reported with its n ; where a stratum holds three or fewer cases we give the per-case values, and proportions carry Wilson 95% intervals.

7 Spinopelvic Baseline

Architecture and training. nnU-Net v2 [Isensee et al., 2020, 2024] ResEnc-M planner, 3d_fullres, nnUNetResEncUNetPlans_100G (100 GB GPU budget). Spacing $0.80 \times 0.75 \times 0.75$ mm, patch $256 \times 320 \times 320$, batch 2; 7-stage residual encoder ($n_{\text{features}}=[32, 64, 128, 256, 320, 320, 320]$, $n_{\text{blocks}}=[1, 3, 4, 6, 6, 6, 6]$, 3^3 kernels, 2^3 strides, instance norm, deep supervision at 5 levels); 9-channel output (bg + 8 fg); ignore-label masked from loss. Class-reweighted DC_and_CE loss ($w=[0.5, 1, 1, 1, 1, 1, 1.5, 2.0, 1, 1]$; last_lumbar boost targets the clinically-important boundary, sacrum boost supports Castellvi precision); SGD Nesterov 0.99, $\lambda=3 \times 10^{-5}$, PolyLR 10^{-2} , 500 epochs (250 train + 50 val iters); 5-fold patient-stratified CV with 6-way LSTV stratification on a training pool of 805 records; 174 records (7 LSTV patients, Table 4) are held out.

Two-tier LSTV emphasis. **Queue-level oversampling:** all five non-normal subtypes duplicated to 25% of per-fold training samples (matching VERIDAH’s $4 \times$ rate). **Class-reweighted CE** (above). No per-patch foreground biasing: under merged labels, biasing toward an L6 class is ill-defined, and biasing the L4/last_lumbar boundary on sac_count would confound the residual-collision measurement.

Hardware. H200 NVL (140 GB HBM3e), 24 CPU workers, preunpacking once per job to a 1.7 TB local-NVMe cache (replacing NFS reads that throttled at 21 MB/s under 5-fold contention). Per-epoch ≈ 400 –700 s; per-fold ≈ 3 days for 500 epochs. Reducing the GPU memory budget forces either smaller patch (loses spinopelvic context) or coarser spacing (loses Castellvi-defining detail); 100 GB is the smallest budget fitting both.

Table 5: Per-class DSC ($\times 100$). **L5** is the unmerged anatomical L5 class; **last_lumbar** is the merged class L5+L6 (identical to L5 on 5-vertebra Normal anatomy). Hip_m merges LH+RH; we train per-side and report both. TotalSegmentator numbers are Normal-subgroup zero-shot on CTSPINOPELVIC1K (Table 1), combined Hips placed in Hip_m. Top-block numbers are reproduced from those works’ own test sets and are not directly comparable; bold marks best in column. [†]Case-weighted mean over the 7 LSTV test patients (per-subgroup in Table 6).

Method	L1	L2	L3	L4	L5	last_lumbar	Sacrum	LH	RH	Hip _m
<i>Reproduced from prior work; not evaluated on CTSPINOPELVIC1K (Li et al. on COLONOG split, FracSegNet on PENGWIN test set)</i>										
3D U-Net	77.19	78.34	81.71	82.86	92.13	—	90.02	93.74	93.42	—
ResNet-50	80.29	76.41	80.62	83.77	91.05	—	88.37	94.72	94.01	—
ResNet-101	74.57	78.11	78.54	83.84	86.35	—	89.05	95.69	94.42	—
SwinUNETR	88.87	83.12	84.36	89.04	92.15	—	91.38	92.33	93.38	—
AttentionUNet	80.26	75.84	79.64	82.32	90.39	—	92.48	93.41	95.45	—
nnU-Net [Isensee et al., 2020]	89.35	91.37	92.41	93.37	95.74	—	96.72	95.50	96.78	—
Li et al. (CFFM+LT) [Li et al., 2025]	92.98	92.36	93.92	93.31	97.07	—	96.98	94.37	96.77	—
FracSegNet [Liu et al., 2025]	—	—	—	—	—	—	98.7	99.3	99.2	—
<i>Reproducible baseline (publicly-released weights), evaluated zero-shot on CTSPINOPELVIC1K</i>										
TotalSegmentator [Wasserthal et al., 2023]	92.1	91.8	92.9	93.6	93.8	—	81.7	—	—	92.9
<i>Ours: CTSPINOPELVIC1K, 5-fold ensemble on held-out test, partial-annotation supervision</i>										
Ours, Normal ($n=167$)	95.3	95.5	95.9	96.7	—	97.1	97.1	95.0	93.6	98.8
Ours, Any-LSTV ($n=7$) [†]	60.1	46.6	31.1	15.8	—	65.2	92.0	97.5	97.6	97.6

Table 6: CTSPINOPELVIC1K 5-fold ensemble on the held-out test split ($n=174$ records), cross-fold mean $\pm$ std. LL EMIT: cases emitting ≥ 100 last_lumbar voxels. Dashes: no GT for that class in the subtype. Semi-sacralization is absent from the test split. Δ vs. TS zero-shot (Table 1). $*n=1$: $\pm$ omitted. [‡]Junction DSC scorable on one case of the subtype; $\pm$ is five checkpoints on that case.

Subgroup	n	L1	L2	L3	L4	last_lumbar	Sac	Hip _m	Jxn DSC	LL emit
Normal	167	.953 $\pm$.008	.955 $\pm$.005	.959 $\pm$.004	.967 $\pm$.005	.971 $\pm$.005	.971 $\pm$.002	.988 $\pm$.001	.907 $\pm$.016 ($n=50$)	167/167
Lumbarization	3	.872 $\pm$.050	.611 $\pm$.123	.377 $\pm$.147	.213 $\pm$.114	.698$\pm$.038	.946 $\pm$.004	.960 $\pm$.002	.946$\pm$.007[‡]	3/3
Sacralization (count)*	1	.988	.959	.734	.187	—	—	—	—	1/1
Sacralization (full)	2	.001 $\pm$.002	.000 $\pm$.000	.001 $\pm$.002	.062 $\pm$.083	.583 $\pm$.136	.910 $\pm$.039	.989 $\pm$.000	.927 $\pm$.068 [‡]	2/2
Ambiguous*	1	—	—	—	—	—	.862	1.000	—	1/1
Δ vs. TS (lumb)	—	−0.6	−24.2	−31.3	−17.5	N/A	+11.3	+3.2	+58.9	—
Δ vs. TS (sacr_count)	—	+72.3	+85.9	+67.9	+18.6	—	—	—	—	—

8 Results

Headline, with its n . The merged-label spinopelvic baseline closes the LSTV junction-DSC gap on lumbarization: junction-DSC rises from TS = 0.357 (3 patients) to ours = **0.946** $\pm$ 0.007 (5-fold mean $\pm$ std; ensemble 0.950), a Δ = **+58.9** $\pm$ 0.7 points. That number is measured on the *one* held-out lumbarization case whose ground truth holds both L5 and sacrum; the other two held-out lumbarization cases are spine-only records, so the $\pm$ is the spread of five fold checkpoints on a single patient, not a population spread. The model emits non-degenerate last_lumbar on all 3/3 held-out lumbarization cases (Wilson 95% interval 44–100%) with per-case ensemble last_lumbar DSC of 0.645, 0.666 and 0.725 (5-fold mean 0.698 $\pm$ 0.038), eliminating the structural absence that drives TS’s level shift. A residual class collision persists on count-style sacralization: on the single held-out sacr_count case, evaluated by all five fold checkpoints (~ 720 K GT-L4 voxel evaluations), **82.7%** of GT-L4 voxels are mis-predicted as last_lumbar (§8.1, Block 2), with per-case L4 DSC of 0.187 $\pm$ 0.337 across folds (one fold’s checkpoint scores mid-range, the others near zero; ensemble 0.032). §8.1 characterizes this as the empirical signature of direct semantic segmentation’s limit on transitional anatomy. On pelvic classes, our model exceeds TotalSegmentator on sacrum by Δ = **+15.2** $\pm$ 0.2 points (TS = 0.817 $\rightarrow$ **0.969** $\pm$ 0.002 5-fold; 0.970 ensemble; 113 held-out records with sacral GT) and establishes state-of-the-art for openly-released dual spinopelvic CT segmentation (Table 5).

LSTV-stratified per-subgroup view. Table 6 exposes the LSTV-induced degradation that aggregate per-class DSC masks; the residual L4 DSC drop on sacr_count cases (§8.1) persists. Semi-sacralization has no held-out case and is therefore unevaluated, not measured; with 7 LSTV test patients the Wilson 95% interval for 7/7 cases emitting last_lumbar is 65–100%.

Table 7: Residual class confusion under merged labels on the held-out split, focused on L4/last_lumbar. Voxel counts aggregated across 5 fold checkpoints; percentages from summed counts. Block 1: GT-last_lumbar on the 3 lumb cases ($\sim 4.06\text{M}$ voxels). Block 2: GT-L4 on the 1 sacr_count case ($\sim 720\text{K}$). Block 3: pred-last_lumbar on that case ($\sim 595\text{K}$).

Block	last_lumbar	sacrum	L4	background
Block 1: last_lumbar GT on lumb ($n=3$)	58.2%	0.4%	40.1%	1.2%
Block 2: L4 GT on sacr_count ($n=1$)	82.7%	0.3%	14.9%	2.2%
Block 3: pred-last_lumbar on sacr_count ($n=1$)	—	0.0%	100.0%	0.0%

8.1 The Residual Class Collision

The merged scheme eliminates the L5/L6 collision driving TS’s level shift. Each fold’s checkpoint evaluates the same 174-case test split; voxel counts below pool 5 evaluations per case (15 case-fold pairs on the 3 lumb cases, 5 on the single sacr_count case). Across 5-fold predictions on lumb cases ($\sim 4.06\text{M}$ aggregated GT-last_lumbar voxels), **58.2%** of GT-last_lumbar voxels are correctly predicted as last_lumbar (Block 1), against TS’s structurally-zero last_lumbar emission. The residual 40.1% misallocation to L4 is the cost of label-merging at the joint, not a re-emergence of TS’s full collapse. The merge does, however, expose a residual collision at the L4/last_lumbar boundary on the sacr_count case: GT-L4 voxels are predicted as last_lumbar in **82.7%** of voxels (Block 2; $\sim 720\text{K}$ GT-L4 voxels aggregated across 5 folds), and predicted-last_lumbar voxels on that case are anatomically L4 in **100.0%** (Block 3). last_lumbar specificity on sacr_count = 0.778 ± 0.074 across folds. Voxels within a case are correlated, so these are dependent observations on one patient: a directional signal that motivates the sequence predictor, not a population estimate.

Same-data unmerged control. To separate the merge from the added data and the added class, we trained the identical network (same planner, patch, loss weights, optimizer, 500 epochs, queue-level LSTV oversampling and the same five folds) on the *unmerged* ten-class scheme (background, L1–L6, sacrum, hips, ignore), with an explicit L6 class and patch-level L6 oversampling switched on. It did not learn L6. In the per-epoch validation confusion block, the share of ground-truth L6 voxels predicted as L5, averaged over the final 100 of 500 epochs, was 83%, 65%, 66% and 97% in the four folds whose validation split holds a lumbarization case (100%, 93%, 65% and 100% at the final epoch; the fifth fold’s validation split holds none), and the validation L6 Dice averaged 0.01–0.13 per fold over the same epochs against 0.91–0.95 for L5. The metric is computed on the validation patches drawn each epoch, so it fluctuates epoch to epoch (29–100% within one fold over the last 100 epochs); the fold-level means are the stable statistic. With the same data, folds and capacity, adding an L6 class does not remove the L5/L6 collision; merging does. This is a limit of dense segmentation at this LSTV sample size (11 lumbarization patients in the training pool), not an impossibility: the L5-versus-L6 cue is a count that exceeds one patch and is data-starved, so counting-aware or two-stage designs, not capacity, are the next test.

Residual upper-lumbar caudal-shift cascade. While the merge restores correct labeling at the bottom-most boundary (Block 1), per-class DSC for upper lumbar on lumb cases shows a residual caudal-shift the merge does not resolve: L1 = 0.872, L2 = 0.611, L3 = 0.377, L4 = 0.213 (Table 6; two of the three cases shift, the third scores L1–L3 above 0.98). TS’s shift concentrates at the bottom (no L6 class); ours distributes it through the upper lumbar, the merge absorbing the bottom shift into last_lumbar—on per-class L2–L4, TS is in fact better, trading upper-lumbar precision for bottom-anchor correctness. Junction-DSC, centered on L5/S1, closes the surgically critical gap (+58.9 pts on the one scorable case) regardless. Resolving the upper cascade requires a sequence predictor with explicit count constraints—contribution (iv): the merge is necessary but insufficient.

Interpretation. The model recovers the merged last_lumbar class on lumbarization, but on count-style sacralization the positional prior dominates: the bottom-most lumbar visible in a patch is treated as last_lumbar regardless of the global count. The disambiguating signal is non-local; a $256 \times 320 \times 320$ patch does not reliably contain both the lumbosacral and the cervico-thoracic boundary needed to count. This confirms the hypothesis behind VERIDAH’s two-stage design: the merge addresses class imbalance, and distinguishing the last lumbar voxels of a 4-vertebra spine from those of a 5-vertebra spine must be deferred to a component with access to global structure.

9 Discussion and Future Work

Generalization and hardware. Any spine segmentation model lacking an L6 class will exhibit the level shift on lumbarization, and §8.1 shows that adding the class at this sample size does not remove it; the LSTV gap is the symptom, not a model-specific bug. Our 100 GB patch budget requires H200-class hardware.

Future work. (1) **VerSe expansion via sacrum pseudolabeling.** VerSe [Sekuboyina et al., 2021] contains 460 CT volumes labeling L6 (label 25) but no pelvis; our segmenter’s sacral DSC (0.969 ± 0.002 , 5-fold) supports sacrum pseudolabeling on VerSe for a 1,613-volume spinopelvic cohort. (2) **Two-stage integration with VERIDAH.** Our segmenter can replace VERIDAH’s VerSe-derived CT upstream, supplying sacrum and per-vertebra centroids in a single forward pass.

Limitations. COLONOG is a US screening cohort (ages 45–75); generalization to emergency/pediatric populations is not evaluated. The six subtypes are functions of the source annotations’ filenames and lumbar counts, not image reads, and the Castellvi grades are a two-resident consensus ($\kappa=0.74$, $n=33$) with no senior read, no blinded re-read and no screening of the normal cohort for Type I; the source-dataset construction biases against morphologically mild Type I presentations (2 of 33), so every quantitative claim is scoped to the conspicuous regime. The held-out test split contains 7 of the 33 LSTV patients (3 lumb, 1 sacr_count, 2 full sacralization, 1 ambiguous, 0 semi-sacralization, which is therefore unevaluated), and junction-DSC is scorable on one lumbarization and one full-sacralization case among them; fold-level evaluation pools five checkpoints over the same 174-case test set, giving 15 case-fold pairs for lumbarization and 5 for sacr_count in the voxel-confusion analysis but only the test-split n for case-level Dice, and the Wilson intervals for strata of 1–7 cases span from 21–65% to 100%. The unmerged control (§8.1) is read from the per-epoch validation confusion block within the cross-validation folds rather than from a full-volume evaluation on the held-out split. TS is evaluated zero-shot, the correct comparison for using TS as a production system but underselling what it could achieve if retrained with an L6 class on far more LSTV data. The wrong-level framing is a mechanism, not a measured outcome: no reader study, planning simulation or workflow evidence is presented.

Implications. A production CT segmenter that emits a confident, well-formed, caudally-shifted prediction on a 5–35%-prevalence variant gives no error signal and, when consumed downstream, reproduces the pattern of the wrong-level miscount. Any spine segmentation model claiming readiness for surgical-planning integration should report per-subgroup DSC and per-class confusion at the L5/S1 junction on a stratified LSTV cohort, with the n of each stratum beside the number.

Data and Code Availability

CTSPINOPELVIC1K is archived on Zenodo under concept DOI 10.5281/zenodo.22139642 (current version v11, 10.5281/zenodo.22797014; see “Release used” in §5 for the April 2026 release the experiments ran on) with core and Responsible AI fields of the Croissant specification [Akhtar et al., 2024]; the CT volumes are mirrored at <https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K>. The five fold checkpoints are at <https://huggingface.co/OpenSpineConsortium/spinopelvic-seg-checkpoints>, trained with <https://github.com/OpenSpineConsortium/spinopelvic-seg> at commit 50b209f; the inference code and a released copy of the trainer are at <https://github.com/OpenSpineConsortium/spinopelvic-seg>, and the data-construction pipeline, the evaluation protocol and the TotalSegmentator benchmark script at <https://github.com/Gregory-Schwing-MD-PhD/CTSpinoPelvic1K>. All are accessible without credentialed access.

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Justification: §5 fully specifies the data-construction pipeline (patient-anchored UID matching, bone-HU placement oracle, Y-axis affine correction, merged-label vocabulary, 6-way LSTV taxonomy, partial-annotation ignore_label protocol). §6 defines the four evaluation metrics. §7 specifies the architecture (nnU-Net v2 ResEnc-M, nnUNetResEncUNetPlans_100G, 3d_fullres), spacing, patch, batch size, full residual-encoder configuration, class-reweighted DC+CE loss with explicit weight vector, optimizer schedule, 500 epochs, two-tier LSTV emphasis, and 5-fold patient-stratified CV with 6-way LSTV stratification. The construction pipeline, trainer source, evaluation code, and trained 5-fold checkpoints are released; see “Data and Code Availability”.

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5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Justification: CTSPINOPELVIC1K is archived on Zenodo under concept DOI 10.5281/zenodo.22139642 (current version v11, 10.5281/zenodo.22797014) with core and Responsible-AI fields of the Croissant specification [Akhtar et al., 2024], and the CT volumes are mirrored on HuggingFace (OpenSpineConsortium/CTSpinoPelvic1K). The experiments were computed on the April 2026 release (§5, “Release used”), whose differences from v11 are stated there. The five fold checkpoints (HuggingFace, OpenSpineConsortium/spinopelvic-seg-checkpoints), the inference code and trainer copy (GitHub, OpenSpineConsortium/spinopelvic-seg), the training repository at the cited commit, and the data-construction pipeline, evaluation protocol and TotalSegmentator benchmark script (GitHub, Gregory-Schwing-MD-PhD/CTSpinoPelvic1K) are public without credentialed access. See “Data and Code Availability”.

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: §7 provides architecture and planner (nnU-Net v2 ResEnc-M, nnUNetResEncUNetPlans_100G, 3d_fullres), spacing $0.80 \times 0.75 \times 0.75$ mm, patch $256 \times 320 \times 320$, batch 2, the 7-stage residual encoder feature/block specification, 9-channel output with ignore-label masking, the explicit class-reweighted DC+CE loss weight vector, SGD Nesterov ($\mu=0.99$, weight decay 3×10^{-5}), PolyLR (10^{-2}), 500 epochs (250 train + 50 val iterations), the two-tier LSTV emphasis (queue oversampling at 25% plus class reweighting), and 5-fold patient-stratified CV with 6-way LSTV stratification. §5 specifies the splits.

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7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: Our segmenter is evaluated under 5-fold patient-stratified CV with 6-way LSTV stratification (§7); per-subgroup DSC are reported as means $\pm$ one standard deviation across the five fold checkpoints evaluated on the same held-out split (Table 6), which captures checkpoint variability, not population variability—the captions say so, and where a stratum holds one to three cases the per-case values are given in the text (§8). Proportions on small strata carry Wilson 95% intervals (closed form; e.g. 3/3 last_lumbar emission: 44–100%; 10/14 lumbarization cases with TS L5 DSC below 0.5: 45–88%). Inter-rater agreement on the Castellvi grades is Cohen’s κ with an asymptotic 95% interval (0.74, 0.55–0.92, $n=33$; a case bootstrap gives 0.53–0.91). TS (Table 1) is evaluated zero-shot once per case; we report per-case means with n per subgroup, the n of scorable junctions, and per-case values for the three-patient lumbarization junction stratum. No normality assumption is made; the unmerged control (§8.1) reports per-fold means over the final 100 epochs with the within-fold epoch range.

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Answer: [Yes]

Justification: §7 (Hardware) specifies H200 NVL (140 GB HBM3e), 24 CPU workers, a 1.7 TB local-NVMe preunpack cache replacing throttled NFS reads, per-epoch $\approx 400\text{--}700$ s, and per-fold ≈ 3 days for 500 epochs (5-fold CV total ≈ 15 GPU-days). §1 documents that the 100 GB GPU-memory budget is the smallest configuration fitting both lumbar spine and pelvis at full spacing; smaller budgets force coarser patches or spacing and degrade LSTV gap closure.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

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Justification: §9 (Implications) addresses both directions. Positive: surfacing a safety-critical, silent failure mode in the most widely deployed CT spine segmenter that produces an output indistinguishable from a wrong-level surgical plan on a 5–35%-prevalence congenital variant,

and providing the LSTV-stratified protocol that any future model claiming surgical-planning readiness must report against. Negative: an LSTV-aware segmenter could itself be over-trusted by downstream pipelines if deployed without radiologist verification; we explicitly recommend that automated outputs require radiologist review for surgical planning, and the paper’s central thesis is precisely that locally well-formed segmentation outputs cannot be assumed safe in the absence of LSTV-stratified evaluation.

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